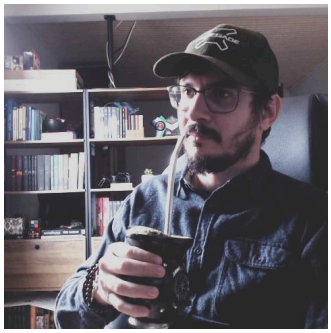


# JOÃO HENRIQUE SERODIO

**Staff Software Architect | Distributed Systems & Backend Engineering** Curitiba, Brazil (Remote) | me@jhserodio.dev | linkedin.com/in/jhserodio | github.com/jhserodio

## EXECUTIVE SUMMARY

**Staff Software Architect** with over 15 years of experience designing and leading **High-Scale Distributed Systems** and **Cloud-Native Architectures** (AWS/Kubernetes). Expert in building robust backend solutions using **Node.js (TypeScript)** and **Java**, with a strong background in migrating legacy monoliths to microservices.

Combines practical engineering leadership with a rigorous academic foundation in **Computational Mathematics**, utilizing **C++** for algorithmic optimization and complex modeling. Proven track record in the **Fintech and Retail sectors** (Stone Co/Pagar.me, Grupo Boticário), translating intricate financial business rules into consistent, fault-tolerant software systems.

## TECHNICAL SKILLS

- **Architecture & Systems:** Distributed Systems, Domain-Driven Design (DDD), Event Sourcing, Microservices, Event-Driven Architecture.
- **Languages: Node.js (TypeScript/JavaScript)**, **Java**, **Rust**, **C++** (Algorithms & Optimization), Python.
- **Data & Persistence:** PostgreSQL (Advanced SQL), DynamoDB (High Scale), MongoDB, ElasticSearch, Redis.
- **Infrastructure:** AWS (EKS, Lambda, Glue, SQS), Kubernetes, Docker, Terraform, IaC.
- **Observability & Messaging:** Apache Kafka, RabbitMQ, SQS/SNS, Datadog, NewRelic.

## PROFESSIONAL EXPERIENCE

**GRUPO BOTICÁRIO** | *Tech Leader (Specialist II) May 2024 – Present* | *Retail Tech*

- **System Design & Consistency:** Architecting the "Suggested Order" engine, a critical B2B financial module. Responsible for the engineering design that ensures accurate calculations for credit limits and inventory allocation based on franchisee financial history.
- **Event-Driven Architecture:** Implemented **Event Sourcing** using **Apache Kafka** to guarantee strict data consistency across distributed services, mitigating risks of financial

discrepancies in high-volume transaction scenarios.
- **Database Optimization:** Designed a polyglot persistence strategy, utilizing **DynamoDB** for high-throughput write operations and **PostgreSQL** for complex relational queries, ensuring optimal performance for financial reporting.
- **Cloud Scalability:** Orchestrated **Kubernetes (EKS)** clusters and **AWS Glue** pipelines to process massive datasets, enabling reliable, near real-time data availability for business operations.

**VNATOR** | *Software Architect & Platform Lead Nov 2018 – Present* | *Technology Solutions*

- **Node.js Architecture:** Leading the backend architecture using **Node.js (NestJS)** and **TypeScript**. Designed scalable microservices capable of handling high-concurrency requests for e-commerce and content platforms.
- **Infrastructure as Code:** Led the modernization of infrastructure using **Terraform** and **CloudFormation**, deploying auto-scaling **Kubernetes** clusters to handle variable traffic loads efficiently.
- **API Design:** Defined standards for REST and GraphQL APIs, ensuring clean contracts and efficient data fetching for frontend consumers.

**PAGAR.ME (STONE CO)** | *Staff Software Engineer Jan 2023 – Jan 2024* | *Fintech / Payments*

- **Financial Data Visualization:** Led the technical architecture and delivery of the Client Finance Dashboard. Focused on the engineering challenges of aggregating and displaying accurate **receivables, fees, and transaction data** in real-time using **React** and **Node.js**.
- **Frontend Architecture (MFE):** Championed the adoption of **Micro-Frontends** (Webpack Module Federation), decoupling engineering teams and accelerating the delivery of financial product features without compromising platform stability.
- **Technical Standards:** Established strict engineering guidelines for the application layer, mentoring senior developers on performance optimization for data-intensive interfaces.

**KOVI** | *Senior Software Engineer Apr 2022 – Dec 2022* | *Mobility / Subscription Model*

- **Serverless Payment Processing:** Developed scalable backend services using **AWS Lambda** (Node.js/NestJS) to manage subscription billing cycles and payment retry logic.
- **System Resilience:** Engineered fault-tolerant payment pipelines using **SQS** and Dead Letter Queues (DLQ) to handle webhook failures, ensuring no transaction data was lost during external gateway outages.

**PEDE PRONTO** | *Senior Software Engineer Jun 2020 – Mar 2022* | *FoodTech / Payments*

- **Microservices Evolution:** Led the extraction of core business logic from a legacy monolith into domain-specific microservices (**Python/Node.js**), improving deployment frequency and system isolation.
- **Automated Onboarding:** Engineered the B2B partner onboarding service, integrating automated background checks and contract generation workflows.

**AGROTIS** | *Software Architect Sept 2016 – May 2019* | *AgTech / SaaS*

- **Multi-Tenant Architecture:** Designed and implemented a secure **multi-tenant SaaS platform**, ensuring strict data isolation between large enterprise clients.
- **Polyglot Development:** Built and maintained critical backend services using **Java** and Python, building a solid foundation in object-oriented design and enterprise patterns.

## EDUCATION

**B.Sc. Computational Applied Mathematics** (In Progress) *Unicentro - Universidade Estadual do Centro-Oeste | 2024 - Present*

- **Focus:** Applying mathematical models (**Graph Theory, Linear Algebra**) to solve complex engineering problems.
- **Algorithmic Solving:** Extensive use of **C++** for implementing high-performance algorithms and solving complex optimization challenges (Data Structures, Computational Geometry).

**B.Sc. Information Systems** *UCP - Faculdades do Centro do Paraná | 2008 - 2012*

## ADDITIONAL INFORMATION

- **Languages:** Portuguese (Native), English (Fluent).
- **Key Soft Skills:** Problem Solving, Technical Leadership, "Disagree and Commit", Adaptability.